TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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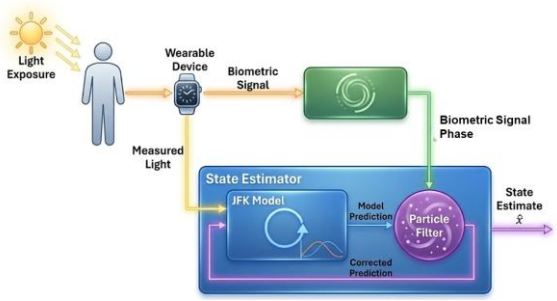
2. Technology Overview

Title of Proposed Technology	The Digital Circadian Twin: Closed-Loop Circadian Regulation Integrating Wearable Biometric Data with Optimal Scheduling of Lighting, Sleep, and Tasks			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Human Resource Competency Development Services	Brain-Cognition-Based Human Performance Enhancement Technologies
Concept of Proposed Technology	An end-to-end platform that converts biometric time series from consumer wearables into individualized circadian-state estimates (circadian phase with quantified uncertainty) and then generates feasible, personalized schedules for light exposure and sleep to improve circadian alignment and productivity. Here, 'digital twin' refers to a personalized, data-updated computational model of an individual's circadian dynamics (state and uncertainty) used to predict responses to candidate light/sleep schedules and to verify system robustness under realistic sensing and adherence conditions. The prototype integrates (i) signal processing and state estimation using particle filtering coupled to a mechanistic circadian oscillator model and (ii) optimization-based planning for scheduling of lighting and sleep under real-world constraints. The platform will also support multi-objective scheduling extensions that incorporate task timing and interventions (e.g., photobiomodulation) as a configurable module, enabling optimization of health- and performance-relevant outcomes such as alertness, fatigue risk, and productivity.			

Rationale for Joint R&D Project	We aim to bring the technology from TRL 3 to TRL 6. This project requires integration with commercial wearable ecosystems, intervention delivery hardware, safety and reliability engineering, user-centered design, and demonstration in pilot-relevant and field environments in Korea. Joint R&D is therefore essential because (a) the estimation-and-optimization core must be co-designed with hardware constraints and deployment realities, and (b) TRL 6 requires a system-level prototype demonstrated in a relevant operational environment. RPI will contribute the scientific and technical foundations: signal processing, state estimation with uncertainty quantification, optimization/control formulations, digital-twin verification, evaluation methodology, and study design/analysis. The Korean SME partner is essential to translate these methods into a deployable product prototype: product-grade software (mobile/UX, cloud backend, wearable data ingestion), lighting/intervention device engineering, safety/quality processes, and Korea-based field deployment and commercialization planning. The Korean SME partner will also be key in adapting the technology to Korean workplace and consumer settings and positioning the technology for downstream certification, manufacturing, and commercialization in Korea. Potential Korean SME partners for this project include ones in the following areas: • Smart lighting / circadian lighting controls (BLE/Wi-Fi/cloud) • Wearable platforms / data SDK integrators • Occupational health / shift-work productivity and safety solutions • Digital health / wellness app companies with user base + delivery channels • Devices that provide other means of circadian and/or sleep intervention, such as photobiomodulation.
Key Words	1. circadian health for jetlag and shift work 2. wearable biometric 3. smart lighting 4. sleep scheduling and intervention 5. alertness

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 3	**End of the Project**	Level 6
R&D Objectives	1. Wearable biometric to circadian state estimation: Develop and validate a robust pipeline that converts consumer wearable biometrics into individualized circadian phase estimates with calibrated uncertainty. 2. Personalized scheduling and closed-loop re-planning: Develop constrained optimization engines that generate feasible, personalized schedules for light exposure and sleep (and optional task timing) and update plans as new wearable data arrive. 3. Deployable integrated prototype: Integrate estimation and planning into a product-grade software prototype with wearable APIs and a controllable lighting/intervention delivery device. 4. Relevant-environment demonstration: Demonstrate usability and effectiveness in a relevant			

	operational environment (e.g., rotating shift workplace pilot study and/or travel re-entrainment pilot study), producing a TRL 6 evidence.
R&D Target Specifications	**<u>Algorithmic performance</u>** A1. Phase estimation accuracy: Median absolute phase error ≤ 1.5–2.0 hours versus reference (e.g., DLMO when available or a validated proxy), with calibrated uncertainty intervals. A2. Robustness to missing data: Maintain target accuracy with ≥ 30% missing wearable samples and common artifacts. A3. Re-planning latency: State update + daily plan generation completed within < 5 minutes (phone/cloud), enabling near-real-time schedule updates. A4. Correctness: ≥ 85% of generated schedules satisfy user constraints (work windows, sleep opportunity, job requirements). **<u>Feasibility</u>** B1. Feasibility: ≥ 70% users report feasibility of following scheduled light/sleep/task windows. B2: Feasibility intervention: ≥ 70% users report feasibility of following scheduled intervention (e.g., photobiomodulation) **<u>System and deployment readiness</u>** C1. Integration: Stable integration with a commercial wearable platforms/API and a controllable lighting/intervention delivery pathway. C2. Reliability: Prototype uptime ≥ 95% during field pilot period; automated handling of missing data and connectivity failures. **<u>Pilot outcome metrics (in relevant environment)</u>** D1. Entrainment benefit: ≥ 20% reduction in time-to-entrainment or equivalent improvement in alignment metrics versus a baseline (generic schedule guidance or standard protocol). D2. User outcomes: Improvement in at least one validated outcome (e.g., subjective sleep quality, fatigue/alertness scale) with pre-specified analysis plan.
R&D Plan	This section presents the detailed research plan to advance the proposed technology from TRL 3 to TRL 6, organized as (1) supporting results, (2) technical approach, and (3) work plan, timeline, and milestones. **1. <u>Supporting Results</u>** This project is supported by a body of completed work that already demonstrates the key scientific and algorithmic building blocks required for a closed-loop circadian regulation system. Our results form an end-to-end chain from state estimation with uncertainty from realistic noisy measurements, using physiologically grounded circadian dynamics models, to optimization-based intervention planning for jetlag and shift-work use cases.  **Fig 1.** Overview of the state estimation framework. **A. End-to-end wearable-to-circadian state estimation with uncertainty quantification** Our baseline approach treats circadian phase as a latent physiological state that must be inferred from noisy measurements and evolving dynamics. This is fundamentally different from (and more deployable than) static regression from wearable features to phase because it naturally supports missing data points, non-stationarity, and individualized parameter adaptation. The end-to-end

structure is captured in Fig. 1, which highlights the essential features of the system: (i) measured light exposure and wearable biometrics act as inputs/outputs, (ii) a mechanistic circadian oscillator model that provides state dynamics, and (iii) a particle filter that updates the belief over circadian state and outputs phase estimates with uncertainty quantification. Uncertainty quantification is essential. In real deployments, biometric signals from wearables are imperfect, e.g., due to loose skin contact, device charging, and travel, potentially leading to imperfect recommendations. A probabilistic state estimator enables conservative, safe recommendations when confidence is low and stronger recommendations when confidence is high. In this project, we will expose state estimation confidence scores to the optimizer, enabling risk management in deployment.

Biometric signal phase: A common barrier to using wearable-based biometric data for circadian state estimation is the need to specify an accurate mathematical mapping from latent circadian state to each observed biometric channel (e.g., how the circadian oscillator state drives heart rate, HRV, or activity traces). Such mappings are difficult to model reliably across individuals and devices. Our framework is designed to bypass this requirement. Rather than attempting to model each biometric time series directly as a nonlinear function of circadian state, we extract an intermediate, physiologically meaningful quantity from the wearable data, i.e., the (relative) phase of the biometric signal. We treat that phase as the observation used for state estimation using widely accepted circadian rhythm models. In effect, we separate the problem into two stages: (i) signal processing that converts raw wearables into a robust phase observation [***Ike2022, Ike2023***], and (ii) particle-filter inference that assimilates this phase observation into a mechanistic circadian oscillator model to estimate circadian phase (and uncertainty). This design makes the system substantially more deployable because it reduces dependence on device-specific amplitude scaling and individual-specific biometric baselines.

Validation using realistic synthetic data: To demonstrate feasibility, we validated the framework using synthetic datasets that replicate key properties of real wearable data streams. Fig. 2 shows our synthetic data generation and test framework. Mechanistic circadian dynamics and realistic noise/perturbations can be used to generate wearable-like time series, including day-to-day variability, measurement noise, and patterns of data dropouts and other randomness typical of consumer devices (charging gaps, intermittent wear, and variable light input availability). This synthetic test capability is a core translational asset. It enables controlled robustness studies (e.g., how data dropouts affect state estimation) and systematic stress tests across parameter variability (e.g., intrinsic period differences).

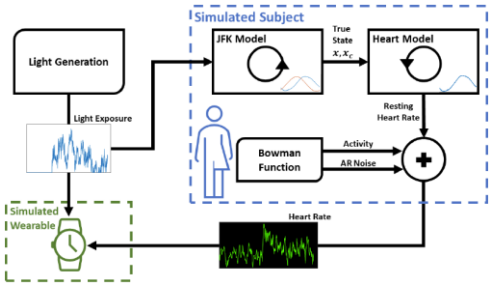


Fig 2. Data generation pipeline. A typical light signal is fed to the JFK circadian model. The resulting state trajectory is passed to the modified electrical heart model that generates a circadian heart rate. Generated activity and an auto-regressive noise process are then added to the basal heart rate to yield the final heart rate. The simulated wearable device is assumed to be able to measure light exposure and heart rate.

Validation using experimental data: Our framework has also been evaluated against experimental datasets collected under controlled protocols that include structured light/sleep schedules and gold-standard circadian phase assessments. In particular, the study design includes DLMO measurements (dim-light melatonin onset) to establish reference circadian phase and phase shift, enabling direct evaluation of whether the inferred circadian state tracks physiologic ground truth. These results demonstrate that combining phase-based observations with particle filtering yields circadian state estimates that are both accurate and robust under realistic sensing constraints. Fig. 3 shows that our method improved alignment between experimentally obtained DLMO data and DLMO predictions derived from circadian state estimation.

	Raw Open-Loop			Conv Open-Loop			Closed-Loop		
Subject	D7 (h)	D14 (h)	Shift (h)	D7 (h)	D14 (h)	Shift (h)	D7 (h)	D14 (h)	Shift (h)
3	0.23	1.25	1.48	0.43	0.40	0.83	0.13	0.57	0.43
4	1.74	1.09	0.65	0.175	0.11	0.28	0.18	0.23	0.05
6	0.91	0.48	0.30	0.01	0.65	0.52	0.18	0.42	0.47
7	1.69	0.57	1.20	0.66	2.68	2.10	0.29	0.63	1.00
8	2.08	3.48	1.40	2.37	3.83	1.47	2.47	3.35	0.88
10	2.52	2.39	0.14	2.94	1.81	1.14	0.42	0.02	0.41
MAE	**1.53**	**1.54**	**0.86**	**1.10**	**1.58**	**1.06**	**0.61**	**0.87**	**0.54**

Fig 3. Errors between collected Dim-Light Melatonin Onset (DLMO) data from our human subject experiment [***Yin2020***] and DLMO predictions from state estimation. Our method (red box) was able to accurately predict with error < 1 hr for most subjects and DLMO shift error within 1 hr for all subjects.

B. Optimization of lighting, sleep, and intervention schedules

Circadian regulation is fundamentally a scheduling and control problem: the same light exposure can either advance or delay circadian phase depending on timing, and the optimal strategy depends on an individual's current phase, constraints, and objectives. Our research therefore focuses on optimization of schedules for lighting, sleep timing, and (when available) interventions under realistic constraints [***Julius2019***, ***Tao2024***, ***Wen2025***]. The optimization objectives are to achieve (i) quickest re-entrainment (e.g., in jet lag recovery), (ii) stabilization under irregular schedules (e.g., in rotating shifts), and (iii) improved alertness/productivity while protecting circadian health.

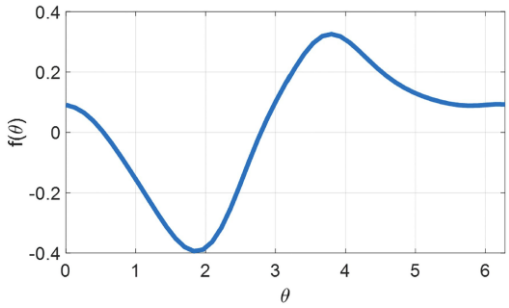


Fig 4. Phase response curve from one of our models. Circadian phase is denoted by θ. Positive PRC $f(\theta)$ indicates phase advancing light, while negative PRC $f(\theta)$ indicates phase delaying light.

Role of models: Our optimal scheduling is grounded in mechanistic circadian models that provide explainability of the solutions. For example, a relatively simple model that involves the phase response curve (PRC) describes why timing matters. See Fig. 4. Applying light at one circadian phase produces circadian phase advance, while light at another phase produces circadian phase delay. This is the rationale for individualized schedules and safety/comfort constraints (e.g., avoid counterproductive late-night bright light when it would delay phase).

Time-optimal re-entrainment for jet lag: Beyond interpretability, our previous work used the models in an optimal control formulation that can compute time-optimal (or near time-optimal) light schedules for large phase shifts such as 8–12-hour jet lag, under intensity and timing constraints. This provides a strong supporting result for the "planner" component of the proposed platform. The algorithm produces explicit schedules that can be delivered via an app or smart lighting system, rather than abstract control signals.

In subsequent work, we extended time-optimal entrainment beyond the circadian clock ("Process C") only formulations by explicitly incorporating sleep homeostasis ("Process S") and the sleep–wake cycle into the control problem. This matters for translation because minimum-time entrainment strategies that ignore sleep can be impractical: they may require excessively long wake periods

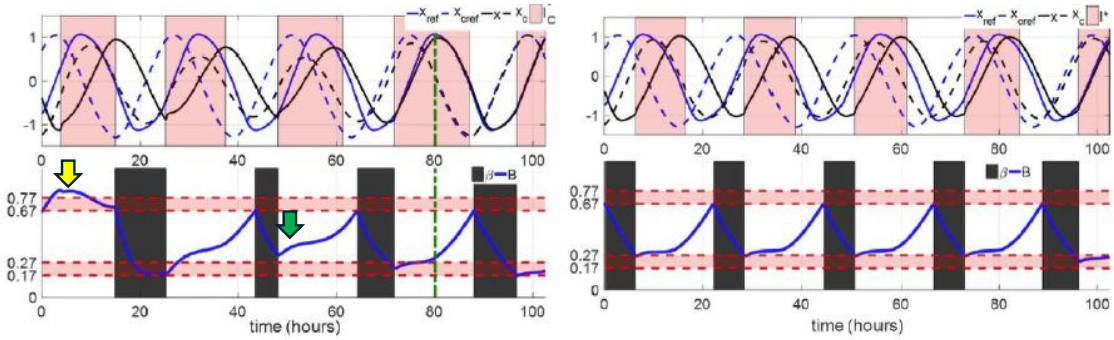


Fig. 5. Optimal lighting schedule to recover from 16-hour jetlag. (Left) without sleep constraint. (Right) with sleep constraint. Top panels show circadian state trajectories. Bottom panels show sleepiness (blue curve) and sleep schedule (black bars). Without addressing the sleep constraint, the optimal lighting schedule causes excessive sleepiness during the first day (yellow bar) and uncomfortable early wake-up time during the third day (green arrow). From [***Yin2021***].

(reported as >20 hours in some cases), compress sleep to 4–5 hours for multiple days, and induce excessive sleepiness. See Fig. 5. In the two-process formulation, we optimize light while enforcing sleep-protection constraints via sleepiness thresholds that govern feasible sleep onset and wake times. The results show that adding sleep scheduling as a decision variable can reduce entrainment time while preserving sleep opportunity, and it avoids excessive sleepiness and unrealistic wake durations seen in Process-C-only entrainment.

Alertness-focused scheduling for shift work: For potential industrial applications, shift work is particularly compelling because outcomes such as alertness and fatigue risk are measurable and closely related to safety and productivity. Our supporting results show that optimally timed bright light can improve alertness trajectories during night shifts [***Yin2022***].

Other potential interventions: The same optimization framework can incorporate additional intervention modalities such as photobiomodulation (PBM) [***Kim2024***]. In this case, PBM can directly impact the sleep dynamics, providing us with an additional control input to be optimized. The optimization objectives can involve one or a combination of the above discussed objectives.

2. Technical Approach

This project advances the current TRL 3 research prototype to a TRL 6, field-demonstrated system by translating the research algorithms into an integrated, deployable platform and validating performance, feasibility, and reliability in Korea-based relevant environments.

Step 1: SME provides deployment context, device/process description, and requirements (foundation for system specifications).
The Korean SME will provide the information needed to define the product and success criteria:
Device/product context: description of the intended product (mobile app and/or lighting/intervention device) and any existing products/modules to be reused.
Wearable integration constraints: targeted wearable platforms and APIs (e.g., preferred vendors), available biometric data (heart rate, heart rate variability, actigraphy, sleep staging, ambient light), sampling/latency characteristics, and data access limitations.
Lighting/intervention pathway: controllable lighting device options, intensity/dimming range, control interface (BLE/Wi-Fi/cloud), safety constraints (maximum intensity, timing constraints), and

deployment constraints (workplace vs consumer setting).
Use-case definition: prioritized scenarios (jet lag, rotating shift work, productivity/alertness scheduling), target cohorts, operational constraints (shift templates, travel patterns), and comparator practices currently used.
Other system requirements: quantitative requirements for reliability/uptime, privacy/security constraints, and any Korea-specific compliance expectations.

Step 2: RPI team adapts and hardens the state estimation and planning core to meet constraints.
Using the SME's specifications, RPI team will translate the research algorithms into components that are directly integrable and testable.
Estimator integration: finalize the particle filter estimator around the wearable signals that are actually available, including phase-based observation extraction, missing data handling, and calibrated uncertainty outputs.
Planner integration: implement constraint-aware scheduling for light and sleep that respects SME device limits and operational constraints, including sleep-protection constraints (sleep opportunity windows, sleepiness bounds) and conservative recommendations under uncertainty. The planner will include an optional task-aware mode that schedules light, sleep, and task timing to optimize an alertness objective derived from the two-process sleep–wake dynamics and low-burden alertness measurements.
Acceptance tests: run in-silico testing of algorithm performance using synthetic data and representative experimental data, define measurable acceptance criteria (accuracy, robustness to data dropouts, constraint satisfaction, runtime) and produce reference implementations and test vectors for SME engineering.

Step 3: Joint integration into a deployable prototype (software + device pathway) and iterative refinement.
RPI and the SME will integrate the estimator and planner with the product stack (mobile/UX + backend + device control) in iterative cycles.
Interface definition and implementation: APIs and data contracts between wearable, state estimator, planner, and schedule delivery.
Monitoring and reliability engineering: instrumentation for uptime, data dropout, recommendation confidence, adherence tracking, and safe fallback behavior.

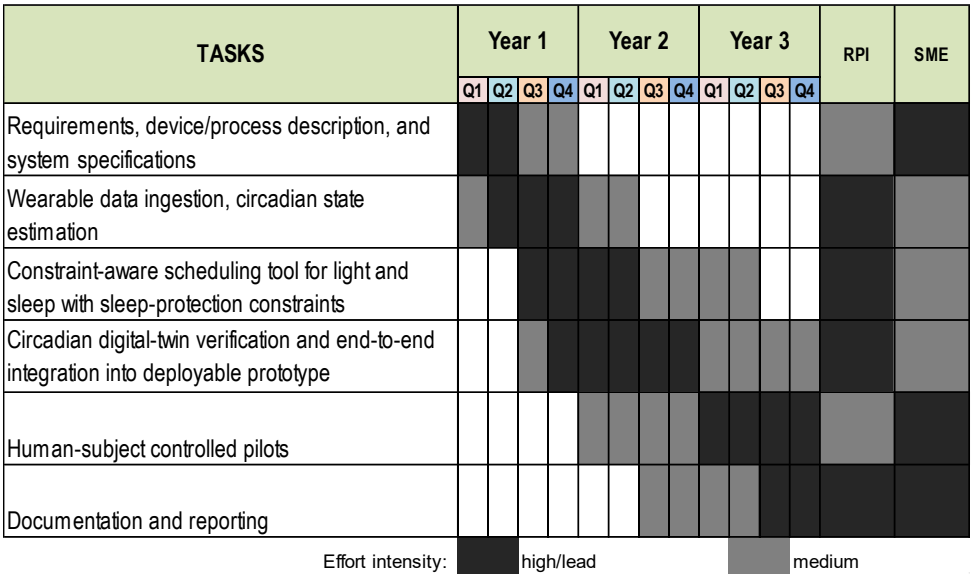
Step 4: Human-subject pilot experiments and field demonstration.
RPI and SME will conduct staged human-subject studies to assess feasibility and measure outcome metrics under managed and then real-world conditions and iteratively refine system design and constraints. RPI will lead study design (protocol, endpoints, analysis plan) and ethics/IRB compliance; the Korean SME will support pilot deployment operations including participant recruitment partnerships in Korea (e.g., shift-work sites and/or travel cohorts), device provisioning (wearables and controllable lighting), onsite/remote technical support, and data-collection logistics. Where feasible, the pilot will include a subset with gold-standard circadian phase reference (e.g., DLMO measurements) to validate phase/phase-shift estimates. The pilots will include secondary/exploratory measures that are short and deployable: brief sleepiness ratings (e.g., Karolinska Sleepiness Scale), a short app-based reaction time task (e.g., a 1–3 minute PVT style test), and task-window adherence logs. These measures support comparative evaluation versus baseline guidance and support product tuning. However, they are not designed as a powered clinical efficacy study.

3. Work Plan, Timeline, and Milestones

The project is planned for 36 months, with six tasks that are interleaved to ensure proper iterative development.
Task 1: Requirements, device/process description, and system specifications

Task 2: Wearable data ingestion, circadian state estimation
Task 3: Constraint-aware scheduling tool for light and sleep with sleep-protection constraints
Task 4: Circadian digital-twin verification and end-to-end integration into deployable prototype
Task 5: Human-subject controlled pilots
Task 6: Documentation and reporting

The Gantt Chart of the overall project can be found below.



Milestones
Month 4: Requirements & specification package completed
Month 9: Biometric signal ingestion and signal processing completed
Month 12: State estimation module completed
Month 18: Schedule optimization module completed.
Month 21: Pilot experiment design and IRB completed.
Month 24: Digital twins verification completed, deployable prototype completed
Month 32: Controlled human-subject pilot complete (incl. DLMO subset when feasible).
Month 36: Final demonstration, software and documentation transfer, submission of report.

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	• Define the scientific/technical requirements for circadian state estimation and schedule generation, aligned to the joint Requirements & Specification Package. • Develop the state estimation functionalities: mechanistic circadian model, particle-filter inference, phase-based observation processing, and uncertainty calibration. • Develop the planner and optimization functionalities: constrained optimization for light and sleep schedules, sleep-protection constraints, and closed-loop re-planning logic. • Define verification and acceptance tests: synthetic stress tests, benchmark datasets, pass/fail criteria, and test data for SME integration. • Lead study design and analysis for human-subject pilots: protocol, statistical analysis plan, and IRB/ethics compliance (as applicable). • Provide reference implementations, integration guidance (APIs/data contracts), and documentation needed for field deployment.

Expected R&R of Korean SME Partner	• Provide device/product description, target use cases, and system constraints; lead the Requirements & Specification Package with RPI review. • Implement the product stack: mobile UX, backend services, data storage, monitoring, and access control. • Integrate wearable platforms/APIs; implement biometric data ingestion and data conditioning/signal processing with RPI's algorithms. • Provide or integrate controllable lighting/intervention pathway: device selection, control interface, safety limits, and companion software. • Operate pilot deployments in Korea: site partnerships, recruitment, device provisioning, participant support, and data-collection logistics. • Support human-subject study operations: scheduling, compliance procedures, and coordination for DLMO sampling when included. • Lead commercialization planning: market selection, partner ecosystem (wearables/lighting), deployment model, and claims strategy.

5. Expected Outcome/Impact

Expected Outcomes and Impact	**Technological achievements** 1. Integrated closed-loop prototype (software + intervention interface): A deployable end-to-end system that performs individualized circadian phase estimation (with uncertainty) from consumer wearables and generates feasible, personalized schedules for light exposure and sleep, integrated with commercial wearable APIs and at least one controllable lighting/intervention delivery device. 2. Validated algorithmic core: Demonstrated estimator accuracy/robustness under realistic missing data and artifacts, and a constraint-aware planning engine that produces actionable schedules aligned with real-world work/sleep constraints. 3. Verification and validation package: Multi-level evidence including benchmarking against theoretical results, digital-twin testing, and relevant-environment pilot/field demonstration. **Commercialization potential** 1. Commercialization pathway for the Korean SME: A prototype architecture that can be packaged as a mobile app and cloud service integrated with major wearable platforms and/or a smart lighting/intervention solution (or partnerships with existing circadian/sleep intervention device makers). 2. Clear initial markets and adoption channels: Jetlag and shift-work solutions for travelers, employers with rotating shift schedules (manufacturing, logistics, healthcare), and wellness or occupational health programs. **Broader industrial and economic impact** 1. Workforce productivity and safety: Better circadian alignment can reduce fatigue-related errors and improve alertness, especially in safety-critical industries that rely on shiftwork (e.g., healthcare, manufacturing and logistics). 2. Growth of Korea's digital health and smart-device ecosystem: The project strengthens Korea-based capabilities in wearable-integrated digital health software, smart-lighting intervention systems, and field-deployment know-how.

6. References

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□ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization